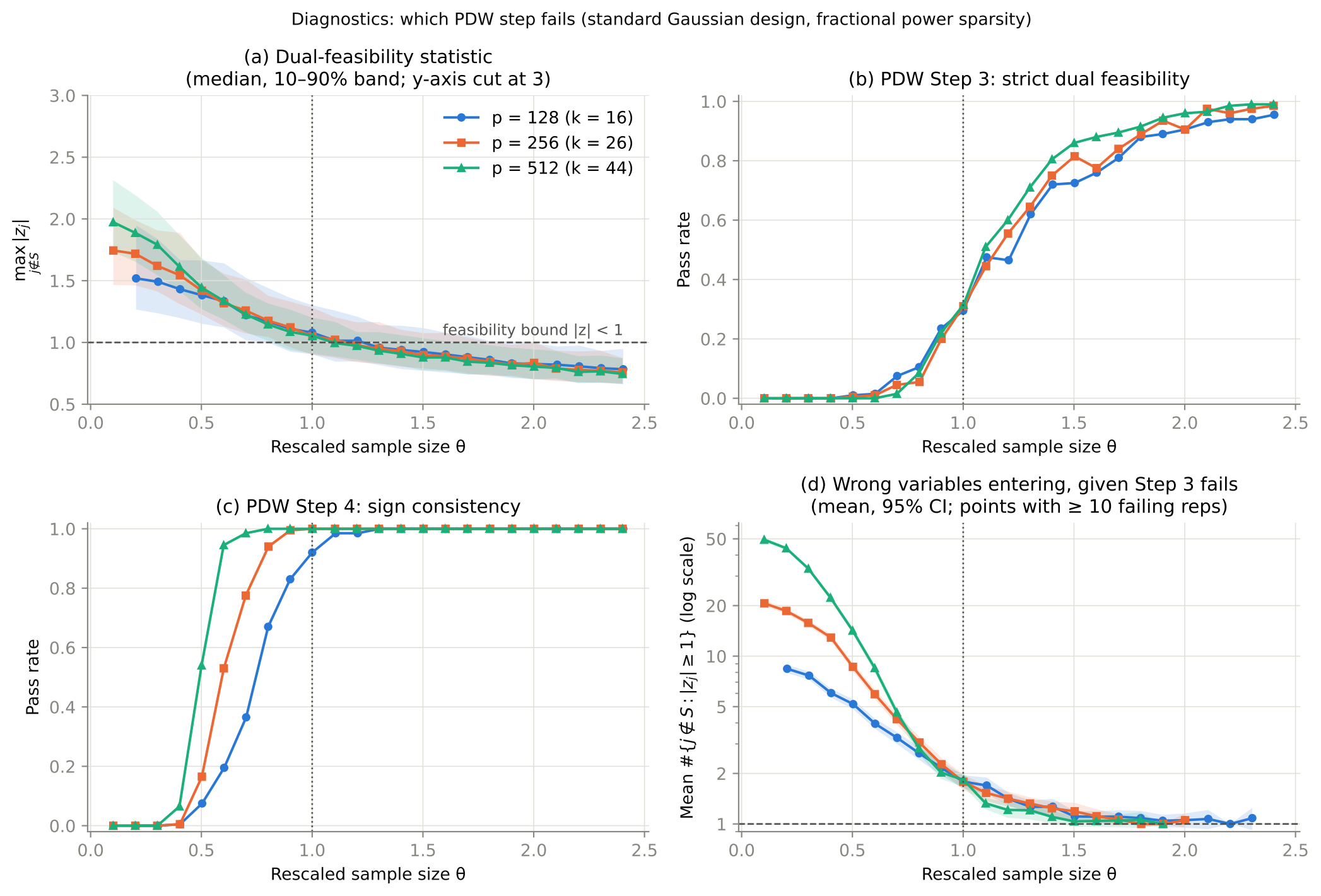
Diagnostics: which PDW step fails (standard Gaussian design, fractional power sparsity)
(a) Dual-feasibility statistic
(median, 10–90% band; y-axis cut at 3)
p = 128 (k = 16)
p = 256 (k = 26)
p = 512 (k = 44)
feasibility bound |z| < 1
max |z_j|
j∉S
Rescaled sample size θ
(b) PDW Step 3: strict dual feasibility
Pass rate
Rescaled sample size θ
(c) PDW Step 4: sign consistency
Pass rate
Rescaled sample size θ
(d) Wrong variables entering, given Step 3 fails
(mean, 95% CI; points with ≥ 10 failing reps)
Mean #{j ∉ S : |z_j| ≥ 1} (log scale)
Rescaled sample size θ